Michael Fischbach

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Malibu, CA

OBJECTIVE

Mechanical Engineering student (Cal Poly Honors, 3.78 GPA) seeking a mechanical design or manufacturing internship. Hands-on CAD-to-part experience across 3-axis CNC machining, prototyping, tolerance analysis, and iterative real-world testing. Has run an independent manufacturing business since 2019, so the design-workshop-testing loop is something owned end to end rather than observed.

EDUCATION

California Polytechnic State University, San Luis Obispo Sep 2025 – May 2029

B.S. Mechanical Engineering, Honors Program | GPA: 3.78 | Dean's List: Fall, Winter, Spring

- Honors research-track program; Learn by Doing curriculum with hands-on manufacturing, CNC machining, and design from first year
- Built multi-component mechanisms (piston, Geneva wheel, CNC-machined screwdriver) with full tolerance stack-up analysis and GD&T

Oaks Christian School – Institute of Engineering Sep 2022 – Jun 2025

Engineering Pathway: 6 multi-month applied design projects | GPA: 3.92 UW / 4.10 W

TECHNICAL SKILLS

CAD / CAE: Fusion 360 (6+ years, 40+ designs), SolidWorks, AutoCAD, Autodesk CFD

Manufacturing: 3-axis HAAS CNC machining, Bridgeport mill, lathe, FDM/SLA 3D printing, carbon-analog composite & concrete casting

Methods: GD&T, tolerance stack-up analysis, DFM, rapid iterative prototyping, design-test-refine, cost modeling

Software: Python, HTML/CSS, Git

Certification: Haas CNC Certification Program – Basic Mill Operator, online coursework completed (Haas Automation, 2026); 12 chapters of CNC safety, setup, and machine operation

Languages: English (native); German (beginner, actively learning)

MANUFACTURING & PROTOTYPING PROJECTS

CNC Milled Golf Putter – Full CAD-to-Part Build 2024 – 2025

3-Axis HAAS CNC, Fusion 360, DFM

- Designed and machined a novel putter geometry from 6061 aluminum on a 3-axis HAAS CNC; ran iterative PLA prototypes to validate geometry before the final aluminum production part
- Performed DFM and cost analysis: modeled warehouse-scale unit cost at $15.78 vs. $50–150 retail
- Selected 1 of 8 finalists from 40+ students; presented the build and process at Capstone Night to 200+ alumni and industry professionals and several hundred students

Wave Energy Converter – Rack & Pinion Generator 2023

Bridgeport Mill, Fusion 360, Fluid Dynamics

- Designed and machined a wave-oscillation-to-electrical-power generator with a five-person team; manufactured all components to tolerance on a Bridgeport mill
- Took 1st place with the highest measured output of three teams: 15 V per motor at 1:12 scale, 360 V scaled
- Applied fluid dynamics and energy-transfer analysis; modeled the full assembly in Fusion 360 and AutoCAD with tolerance documentation

Chameleon Ramps – Product Design & Manufacturing 2019 – Present

Fusion 360, Casting, Iterative Prototyping *chameleonramps.com*

- 6-year iterative development program: designed 40+ unique products and developed proprietary casting techniques, reducing material waste ~30%
- Ran a systematic design-test-refine loop across 4–6 prototype generations per product; 1358 orders fulfilled and $50,000+ in revenue

SMC Trading Bot – Autonomous Options Execution 2025 – Present

Python, Options & Futures, Broker APIs, SQLite, Backtesting

- Built an autonomous Python system that parses unstructured options alerts and places live orders against a funded Webull brokerage account, with risk limits, position sizing, and state reconciled from broker fills; a second lane trades futures on Tradovate
- A misread alert becomes a real order, so every parse and order path is pinned by a regression suite of 2,015 tests before it reaches the live account
- Selected strategy parameters on their worst case across two disjoint validation windows rather than their best, using a custom backtesting harness: the best-fit choice is how a system that backtests well loses money live

EXPERIENCE

CEO & Founder Jun 2026 – Present

Campus Native *San Luis Obispo, CA*

- Founded and ran a digital marketplace; recruited and led a 4-person intern team across engineering, marketing, and content

Data Analyst Mar 2023 – Present

All Domain Integration (ADI) – Aerospace Engineering *Remote*

- **Active Secret security clearance**: granted for work on a classified contract
- 3+ year engagement with an aerospace engineering firm: track and report web traffic and user-behavior data in Google Analytics, and turn that data into content and search-visibility decisions

Business Partner May 2023 – Jul 2024

Ashersells LLC *Malibu, CA*

- Helped scale operations to $500,000+ annual revenue; built Python automation for logistics and order processing

LEADERSHIP & INTERNATIONAL PROFILE

- **Cal Poly Honors Program** – Admitted Fall 2026; selective research-focused program with faculty mentorship
- **Eagle Scout** – Boy Scouts of America (2025); Eagle rank, Senior Patrol Leader 2 years; 15 years leadership
- **Dean's List** – Cal Poly SLO, all three quarters (Fall / Winter / Spring 2025–26)
- **1st Place, Noodle Bridge Competition** – won by a 38.5% margin
- **Internationally Mobile** – U.S. Air Force family; 9 relocations across 7 states and territories; adapts quickly to new environments and teams
- **Community Service** – 4+ years bimonthly meal service to unhoused community; 2021–present